It is convenient to represent the above differences in a table as shown below.

Forward Difference Table for y :

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
x_0	y_0					
		Δy_0				
x_1	y_1		$\Delta^2 y_0$			
		Δy_1	ASS	$\Delta^3 y_0$		
x_2	y_2		$\Delta^2 y_1$		$\Delta^4 y_0$	
		Δy_2		$\Delta^3 y_1$		$\Delta^5 y_0$
x_3	y_3		$\Delta^2 y_2$		$\Delta^4 y_1$	
		Δy_3		$\Delta^3 y_2$		
x_4	y_4		$\Delta^2 y_3$			
		Δy_4				
x_5	y_5					

The forward difference table for $f(x)$ is given below.

x	$f(x)$				
		$\Delta f(x)$			
$x+h$	$f(x+h)$		$\Delta^2 f(x)$		
		$\Delta f(x+h)$		$\Delta^3 f(x)$	
$x+2h$	$f(x+2h)$		$\Delta^2 f(x+h)$		$\Delta^4 f(x)$
		$\Delta f(x+2h)$		$\Delta^3 f(x+h)$	
$x+3h$	$f(x+3h)$		$\Delta^2 f(x+2h)$		
		$\Delta f(x+3h)$			
$x+4h$	$f(x+4h)$				

Backward Difference operator (∇) :

Let $y = f(x)$ be a given function of x. Let $y_0, y_1, \ldots, y_n$ be the values of y at

$x = x_0, x_1, x_2, \ldots, x_n$ respectively. Then

$$y_1 - y_0 = \nabla y_1$$

$$y_2 - y_1 = \nabla y_2$$

$$y_n - y_{n-1} = \nabla y_n$$

are called the first(backward) differences.

The operator ∇ is called backward difference operator and pronounced as nepla.

Second (backward) differences:

$\nabla^2 y_n = \nabla y_n - \nabla y_{n+1}$, $n = 1,2,3,\ldots$

XII Std - Business Maths & Stat EM Chapter 5.indd 106 2/28/2020 6:08:29 PM